Algorithmic Problem Solving	**M1 MOSIG**

Flows and Cuts

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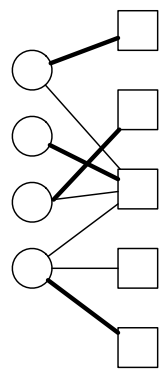
1 Maximum Cardinality Bipartite Matching

You are the father/mother of a very large family and you want to give gifts to your n children. You have m gifts available. You know that certain gifts are not suitable for certain children. Concretely, you are given a bipartite graph $G(U, V, E)$ with $|U| = n$, $|V| = m$, and $(u, v) \in E$ if gift v is suitable for child u. You want to find a maximum *matching* in this graph, which is defined as the largest set $M \subseteq E$ such that each vertex of the graph is incident to at most one edge in M.



Reduce this problem to a maximum flow problem.

2 Baseball Elimination

We are in the middle of a baseball competition. I do not know much about this game, except that:

- Games are scheduled in advance.
- In each game exactly one of the two teams wins.
- At the end of the tournament each team has won a certain number of games. Those for which this number is not maximal are eliminated.

In the example below, Montreal will be eliminated, because even if it wins each of its remaining 3 games, it would reach only 80 wins, and Atlanta has already won more than that. Philly could reach a score of 83, but if Atlanta loses all its games, others will win instead, and ultimately surpass Philly.
Given n teams, the number w_i of games already won by each team i, and the list of remaining games (i, j), we want to determine whether a particular team k still has a chance to win.

team	wins	losses	to play	against $= r_{ij}$			
i	w_i	l_i	r_i	Atl	Phi	NY	Mon
Atlanta	83	71	8	–	1	6	1
Philly	80	79	3	1	–	0	2
New York	78	78	6	6	0	–	0
Montreal	77	82	3	1	2	0	–

Find a reduction of this problem to a maximum flow problem. Hint: the graph you will construct will resemble that of the previous exercise, but with different capacities.

3 Entrepreneur

You are a entrepreneur who has received a contract to construct a building. This work decomposes into n tasks, numbered from 1 to n. These tasks are linked by a precedence order $\prec$, where $i \prec j$ means that task j can only be performed once i has been completed. You will be paid gradually as each task is completed. Task i earns you w_i Euros, but costs you p_i Euros to perform.

You do not intend to perform all the tasks, only a subset $S \subseteq \{1, \ldots, n\}$ with the property that if $i \prec j$ and $j \in S$ then $i \in S$ as well. Such a set is called an *initial* set for the order $\prec$. Your goal is to find an initial set S that maximizes the ratio

$$\frac{\sum_{i \in S} w_i}{\sum_{i \in S} p_i}.$$

Find a reduction of this problem to a maximum flow problem. Hint: try determining whether there exists a solution with ratio at least R. Afterwards, you can perform a binary search on R.